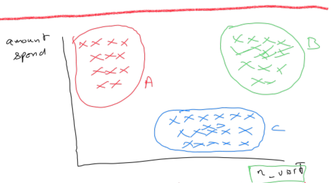
Clustering

[y] ⇒ target → [model] ⇒ supervised

↳ un-supervised ⇒ NO Target

data points that are close will be similar

[Judgement] ⇒ Business Use Case



A → Impulsive spender, rich people

B → Rich, upper-middle class, heavy shopper

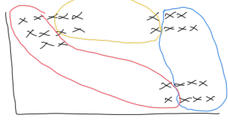
C → lower middle class, window shopper

K means

↳ # of clusters

↳ Hyper Parameter

D {x^i, $x^i \in R^n$} X no Y
no tags



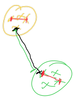
→ clusters should be very separated (inter-cluster dist ↑↑)

→ data points in the same cluster should be close (intra cluster dist ↓↓)

Dunn Index = $\frac{\text{inter cluster}}{\text{intra cluster}}$

min (dist b/w the points of 2 different clusters) ↑↑

max (dist b/w the points of same cluster) ↓↓

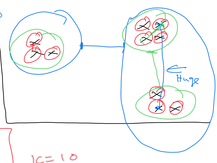


total # = 10

$I = \frac{0}{2} = 0$

$Z = \frac{\text{Inter dist}}{\text{Intra dist}}$ ↑↑

= very small



$K=3 = \frac{\text{Inter dist}}{\text{Intra dist}}$ K = 10

⇒ Inter dist ↓↓ ⇒ 0

Intra dist = 0 ⇒ + small

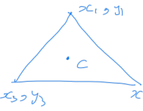
↳ ⇒ 0

0 → Infinity

↳ Dunn Index

Lloyd's Algo (K means)

Centroid



$C_{(x)} = \frac{x_1 + x_2 + x_3}{3}$

$C_{(y)} = \frac{y_1 + y_2 + y_3}{3}$

Lloyd's Algo (k means)

↳ Hyper parameter

⇒ K (should be decided)

① initiate k random centroids for each of k clusters (pick random data points)

② For each & every data point, find the closest centroid

↳ How to find?

③ Find new centroids

↳ based on the data points where they were the closest

↳ go to every dimension

← take average

mean k means

④ keep on repeating ② ← ③

Euclidean dist (low dimension)

Man hattan dist (medium # dimensions)

Cosine dist (High dim)

Inertia $\sum_{k} \sum_{\text{every point in the cluster}} (\text{point} - \text{centroid})$

$\sum_{k} \sum \text{dist}(P_1, C_k) + \text{dist}(P_2 + C_k) \ldots$

$\sqrt{(P_1 - C)^2}$

WCSS (within cluster sum of sqrs) ⇐⇒ $(P_1 - C)^2$